

UPI Transaction Analysis Dashboard – Executive Report

Capstone Project | Data Analytics With GENAI

1. Executive Objective

The objective of this analysis is to evaluate UPI transaction performance, identify fraud patterns, analyze failed transaction trends, and assess customer churn behavior using Power BI dashboards. The report delivers data-driven insights and strategic recommendations to improve transaction reliability, fraud prevention, and customer retention.

2. Key Business Metrics

Metric	Value
Total Transactions	100K
Total Transaction Amount	4.24M
Success Rate	92.14%
Failure Rate	5.87%
Pending Rate	1.99%
Total Fraud Transactions	2,000
Fraud Rate	2.00%
Total Failed Transactions	5,871
Churned Customers	1,337
Active Customer Rate	7032

3. Top 5 Data-Driven Insights

- **Insight 1:** UPI transaction volume demonstrated exponential growth between 2023 and 2025, indicating rapid digital payment adoption across regions and banking platforms.
- **Insight 2:** Fraud activity increased significantly in 2025, with users above 50 years identified as the most vulnerable customer segment.
- **Insight 3:** QR-code-based transactions contributed the highest fraud distribution among all transaction channels, highlighting elevated security risks.
- **Insight 4:** Android devices recorded the highest failed transaction count, suggesting device-specific performance or network reliability issues.
- **Insight 5:** Customer churn analysis revealed that 19.01% of users became inactive, while central and western regions showed the highest churn concentration.

4. Supporting Dashboard Visualizations

- Transaction Growth Trend (2020–2025)
- Fraud Activity Trend by Month and Year
- Device-Wise Failure Distribution
- Regional Merchant & Customer Churn Distribution
- Payment Status and Customer Segmentation Analysis

5. Strategic Recommendations

1. **Strengthen Fraud Detection Systems:** Implement AI-driven fraud monitoring for QR-code transactions and high-risk customer segments to reduce fraudulent activities.
2. **Improve Device & Transaction Reliability:** Conduct performance optimization and compatibility testing for Android devices to reduce failed transaction rates.
3. **Launch Customer Retention Programs:** Introduce personalized cashback offers, loyalty rewards, and targeted engagement campaigns for inactive and at-risk customers.
4. **Enhance Customer Awareness:** Run digital payment safety campaigns focusing on elderly users to minimize phishing and fraud-related risks.
5. **Regional Performance Monitoring:** Develop region-specific operational strategies to address churn concentration and improve merchant engagement.

6. Conclusion

The Power BI dashboard successfully provides a comprehensive overview of UPI transaction performance, fraud trends, failed transactions, and customer churn behavior. While the overall transaction success rate remains strong at 92.14%, increasing fraud incidents and transaction failures indicate the need for stronger security frameworks and infrastructure optimization. Strategic implementation of AI-based fraud detection, customer engagement initiatives, and operational monitoring can significantly improve digital payment efficiency and customer trust.

7. Tools & Technologies Used

- Power BI
- Python (Pandas & NumPy)
- MySQL
- ODBC Connector
- DAX Calculations
- Excel / CSV Data Sources
- Data Cleaning & Transformation
- Business Intelligence & Dashboard Design